1.1.2 Databases: Dynamic sites need data

Views are key to the *dynamic* part of a dynamic website. They can create what is on a page based on data or input from the user. That data might be products in a catalog, the score of last night's game, or your latest recipe for gnocchi. In all these cases, your data has to live somewhere. Enter the database.

Databases are a field in computing in their own right. There are different kinds, each with its advantages and disadvantages. There are many implementations of each kind and specialty languages written just for communicating with them. All of this can be a little overwhelming. Django implements an ORM to abstract this away.

Consider a database containing information about books, like the one shown in figure 1.6. You are going to have different kinds of data: an author (that's me), a publisher (that's Manning), and a book (that's what you're holding). There are many authors, many books, and many publishers. Some books have multiple authors, and some authors have written multiple books. A relational database stores this information in tables similar to a sheet in an Excel spreadsheet. It then expresses relationships between these items with keys. The Author table might reference the Book table using a key that is a unique identifier for a Book.

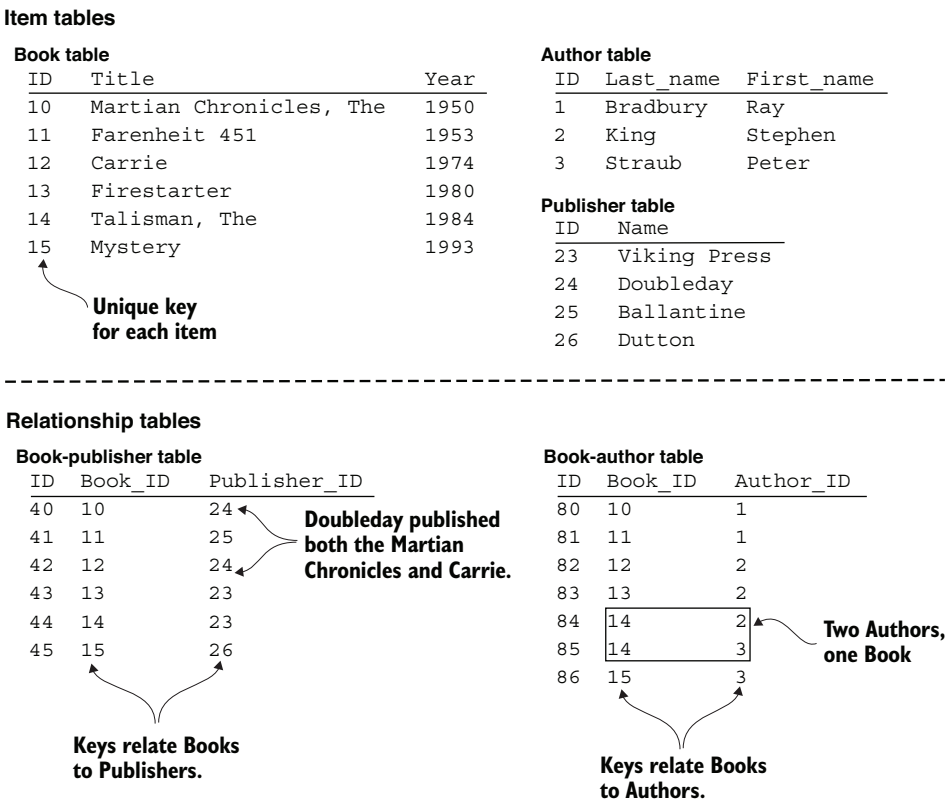


Figure 1.6 Tables store either item information or relationship information